

Compound Interest

LIVE CLASS

Zero to Hero Level

CONCEPT SHEET

Imp:

C.I = Successive change
$= a + b + \frac{ab}{100}$

C.I

Formula (1)

$A = 1000\left[1 + \frac{10}{100}\right]^2$

$A = 1000 \times \frac{110}{100} \times \frac{110}{100}$

$A = [\quad]$

$C.I = A - P$

P = 1000
r = 10% p.a
t = 2yr

II (C.I/S.I)

$10 + 10 + \frac{100}{100}$

21%

$\frac{21}{100} \times 1000$ Ans

C.I = 210

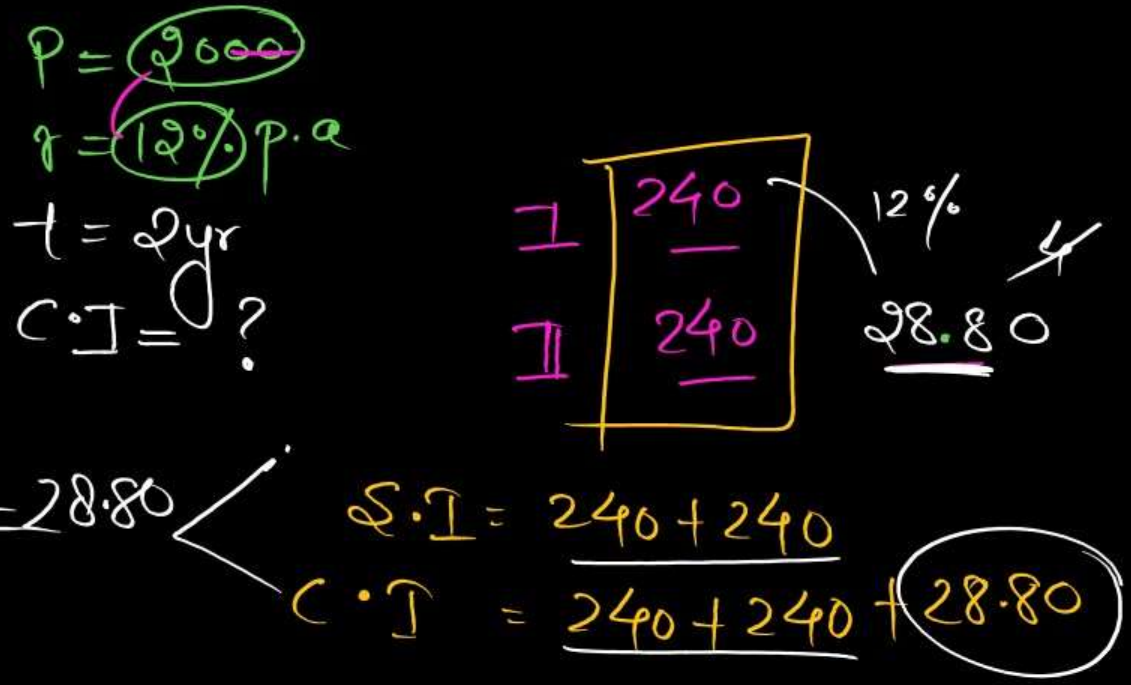
Imp (fast)

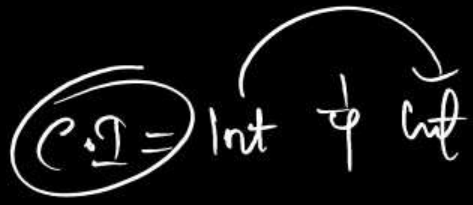
I 100 → 100 r = 10% → 10

II

210

CONCEPT SHEET

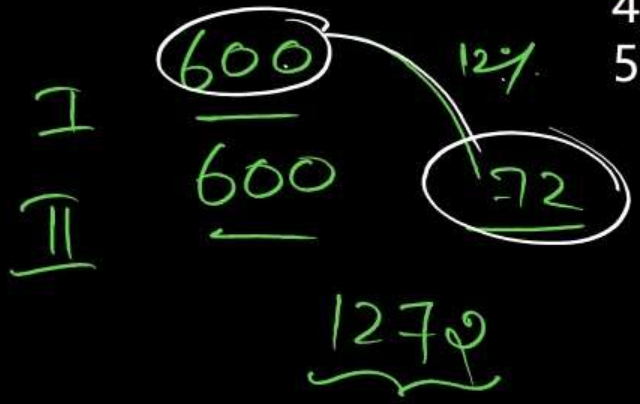




Question1 :

What will be the compound interest on 5000 for 2 years at 12% per annum?

1) 1250
2) 1200
3) 1272
4) 2174
5) None of these



Question 2:

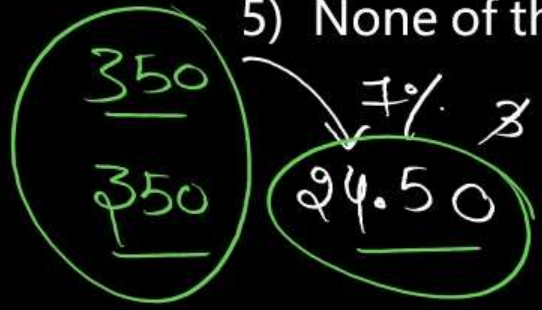
.What will be the compound interest on 5000 for 2 years at 7% per annum?

1) 725
2) 700
3) 724.50
4) 714.50
5) None of these

S.I = 700

C.I = 724.50

D =



Question 3:

Amit Kumar invested an amount of Rs 15000 at compound interest rate of 10 pcpa for a period of 2 years. What amount will he receive at the end of two years?

18150 Ans

TITA

1500

1500 150

C.I = ₹150

$A = P + Int$

$15000 + 3150$

Question 4:

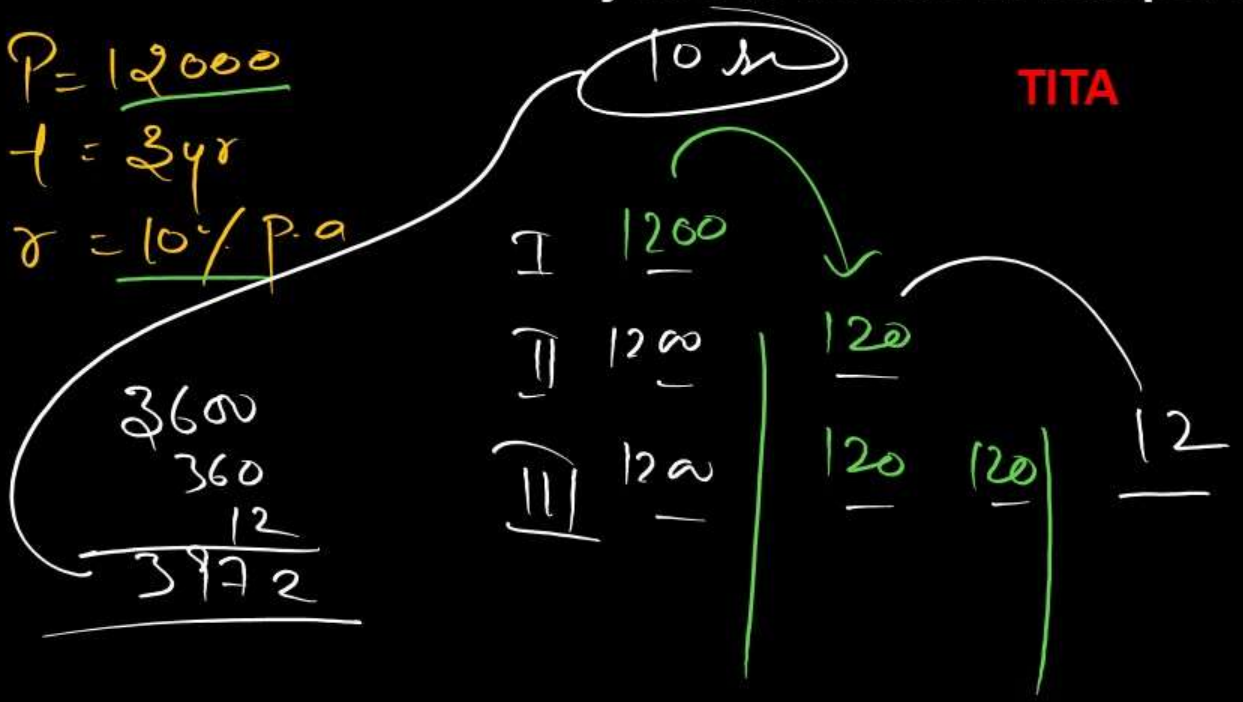
Raviraj invested an amount of Rs 10,000 at compound interest rate of 10 per cent per annum for a period of three years. How much amount will Raviraj get after 3 years?

TITA

Question 5:

What will be the compound interest acquired on a sum of Rs 12000 for 3 years at the rate of 10% per annum?

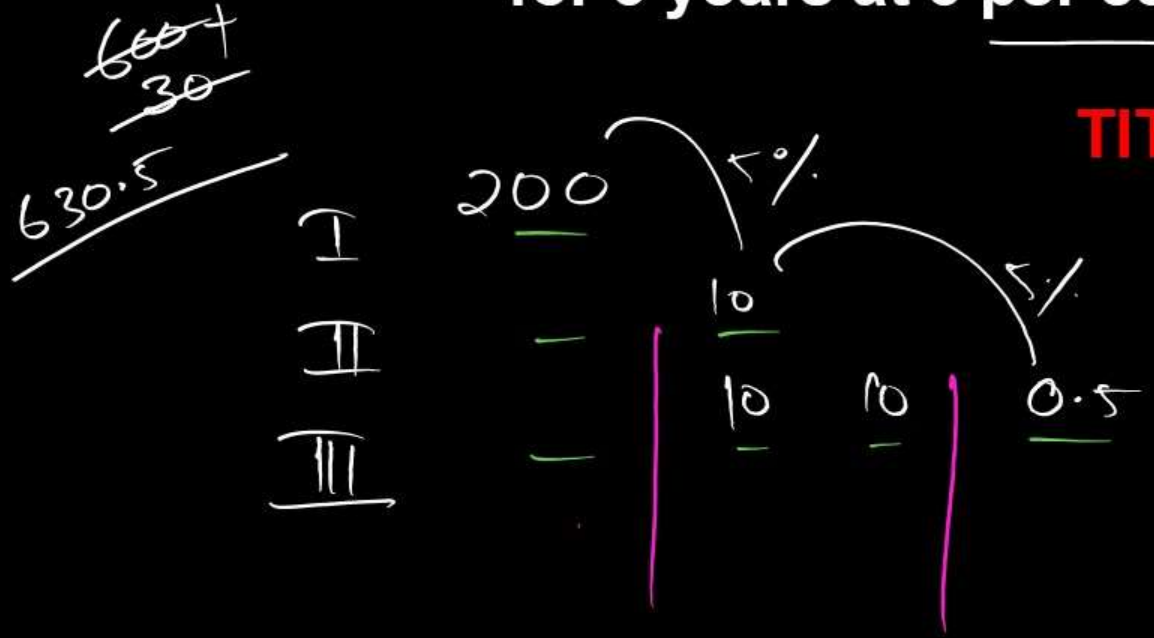
TITA



Question 6:

Find the compound interest on Rs 4000 for 3 years at 5 per cent per annum.

TITA



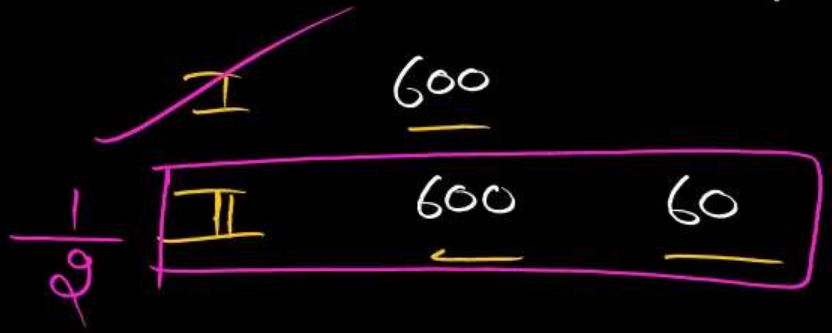
1 yr 4 months = 1/3

Question 7:

The Compound Interest on Rs 6000 at 10% p.a for $1\frac{1}{2}$ yr when the interest is being compounded annually is ?

930 Ans

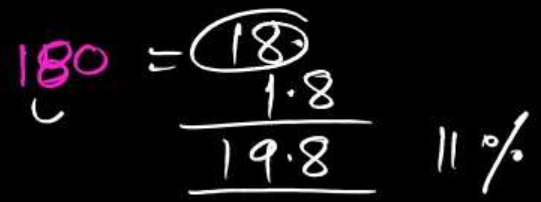
TITA

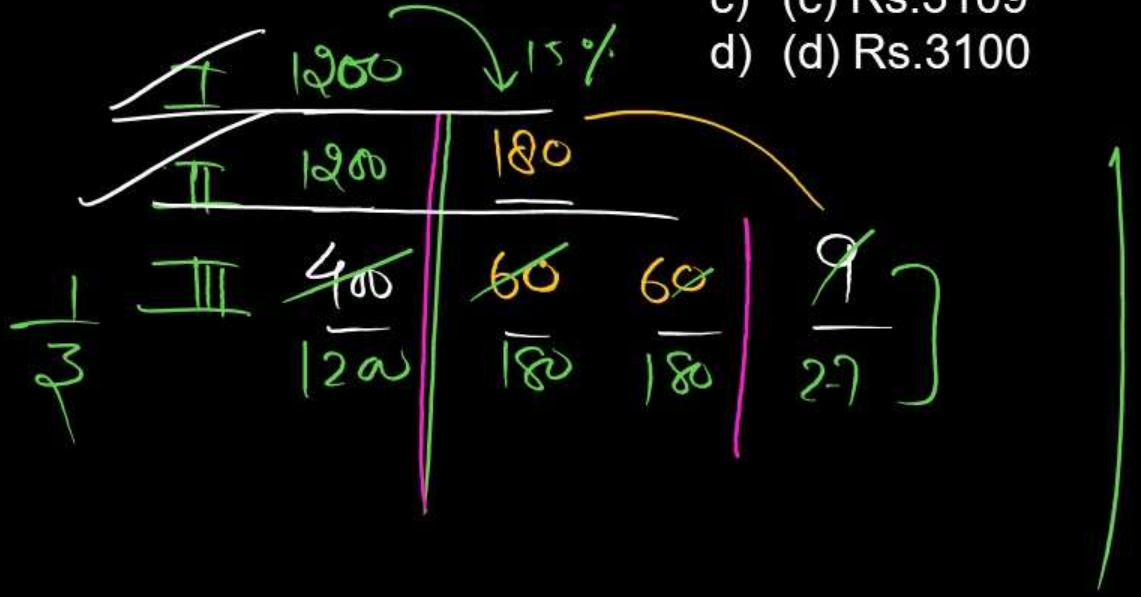


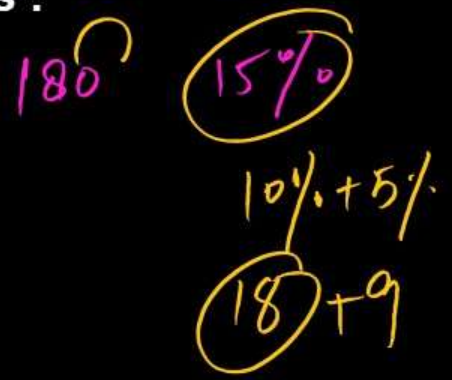
Question 8 :

The compound interest on Rs.8,000 at 15% per annum for 2 years 4 months, compounded annually is :

a) a. Rs.2980
b) (b) Rs.3091
c) (c) Rs.3109
d) (d) Rs.3100



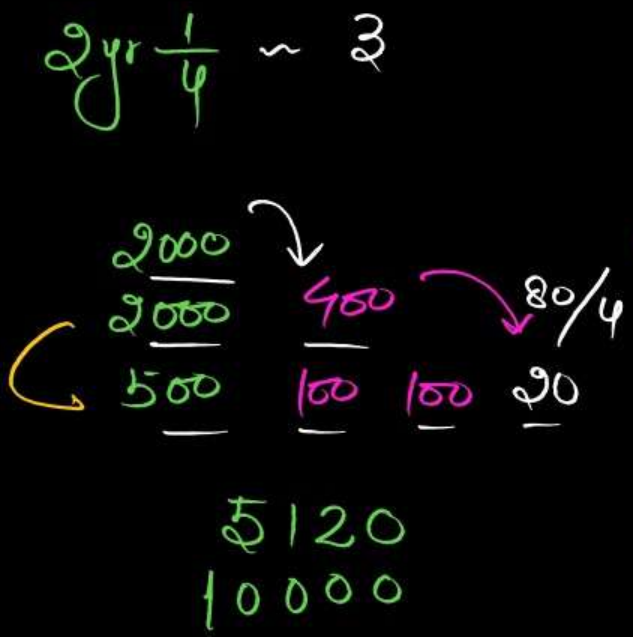


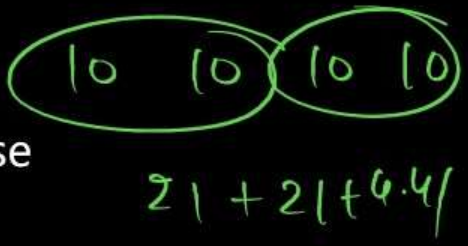


Question 9:

An amount of 10000 was deposited in a bank for a period of 27 months at the rate of 20% pa on compound interest. What will be the amount received on maturity?

1) 15120
2) 12400
3) 14260
4) 12500
5) None of these





Question 10:

Amit has given a loan to Sumit an amount of 20000 at an interest rate 8 % p.a. for a period of 30 months. If interest charged is at compound interest, how much does Sumit need to pay in the end?

1) 23000
2) 2) 24000.36
3) 3) 24261.12
4) 4) 25020.54
5) 5) None of these

$2 yr \frac{1}{2}$

Question 11:

A sum of 40000 is invested for 18 months at 20% p.a. on compound interest. If the interest is compounded half yearly, what will be the interest to be paid?

1) 13530
2) 13080
3) 13540
4) 13240
5) None of these

$P = 40,000$

$t = 3$

$r = \frac{20}{2} \times 6 = 10\%$

4000

400

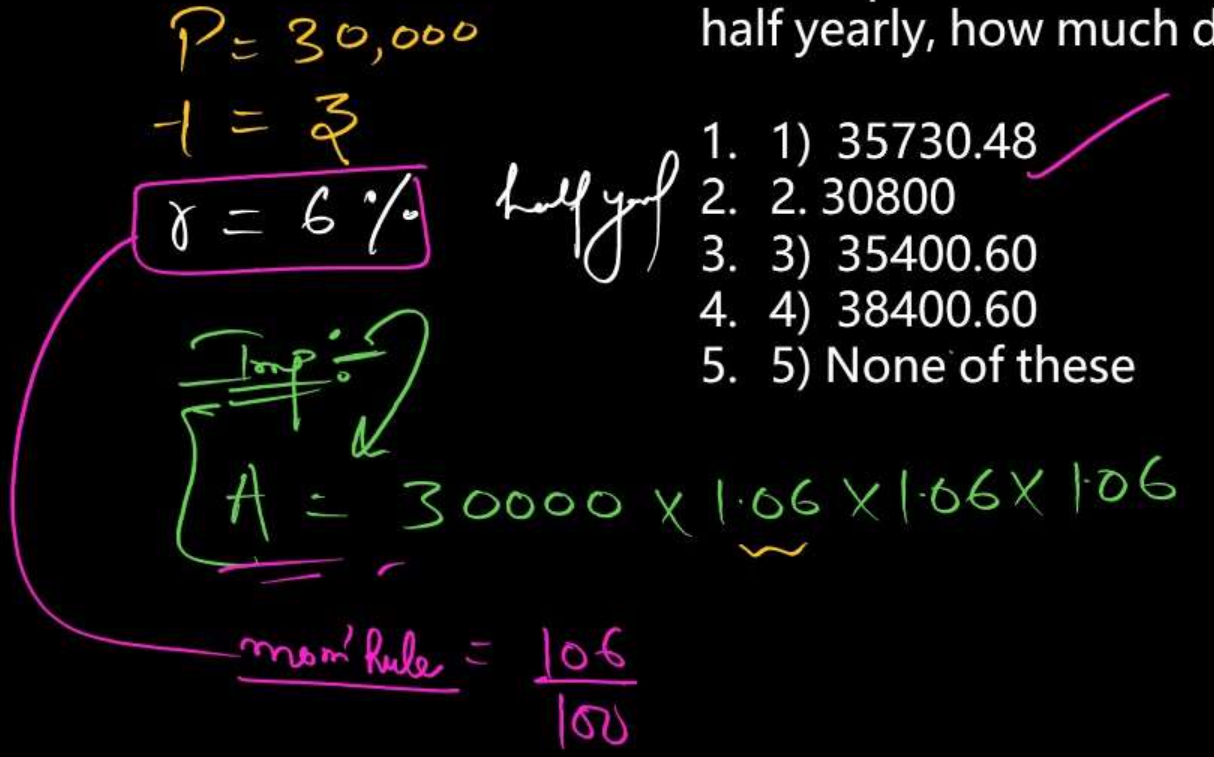
40

12000 + 1200 + 40

Question 12:

A sum of 30000 is invested for 18 months at 12% p.a. on compound interest. If the interest is compounded half yearly, how much does it become on maturity?

1. 1) 35730.48
2. 2. 30800
3. 3) 35400.60
4. 4) 38400.60
5. 5) None of these



Question 13:

The Compound interest on Rs.16,000 for 9 months at 20% per annum. Interest being compounded quarterly, is

1. **Rs.2,520 (b)Rs.2,524(c)Rs.2,522 (d) Rs.2,518**

$\frac{20}{12} \times 3$

$P = 16000$

$t = 3$

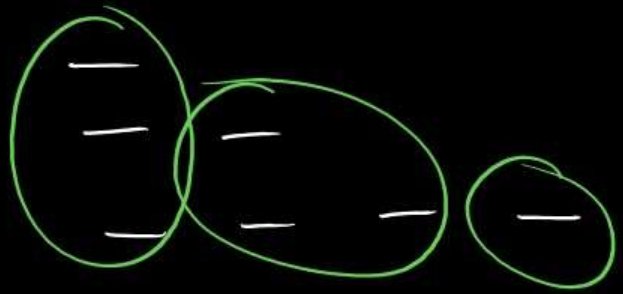
$r = 5\%$

$800 \times 3 = 2400$

$40 \times 3 = 120$

$2 \times 1 = 2$

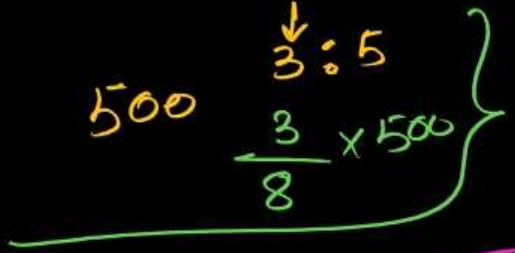
2522

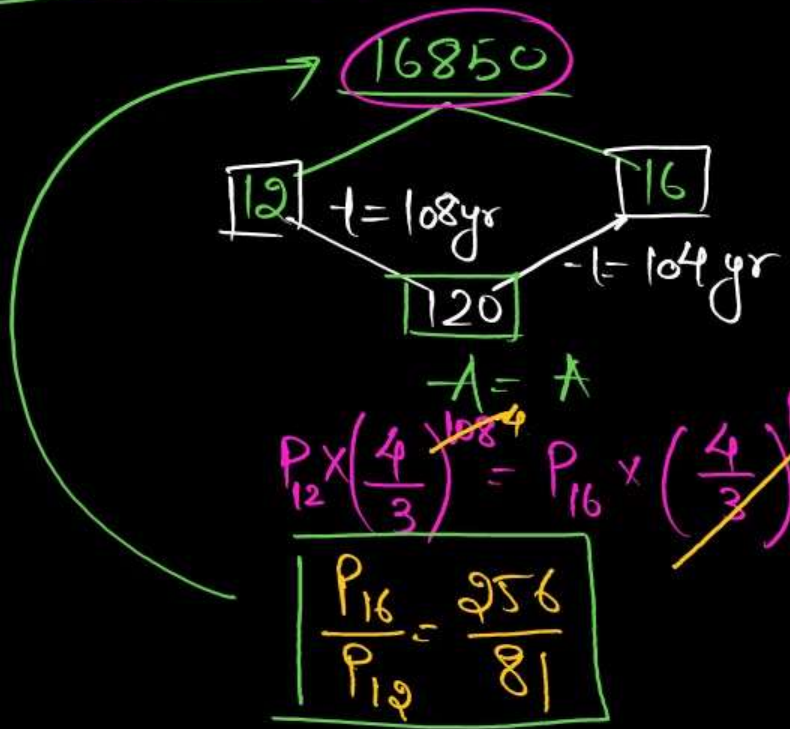


Question 14:

A man wants to invest Rs. 16850 in bank account of his 2 sons, whose ages are 12 years & 16 years in such a way so that they will get equal amount at an age of 120 years at the rate of $33\frac{1}{3}\%$ compounded annually. Find the share of younger son?

(a) 4050 (b) 4080 (c) 4090 (d) 4085





Question 15:

A sum of money invested at compound interest amounts to 800 in 3 years and 882 in 5 years. What is the rate of interest? 1) 2.5% 2) 4% 3) 5% 4) 6.66% 5) None of these

$\sqrt[2]{\frac{441}{400}} = \left(1+\frac{r}{100}\right)$

$\frac{21}{20} = \left(1+\frac{r}{100}\right)$

$\frac{21}{20} = \frac{100+r}{100}$

$105 - 100 = r$

$r = 5\%$

$A = P\left(1+\frac{r}{100}\right)^n$

$882 = P\left(1+\frac{r}{100}\right)^5$

$800 = P\left(1+\frac{r}{100}\right)^3$

$\frac{882}{800} = \frac{441}{400} = \left(1+\frac{r}{100}\right)^2$

Question 15(B):

A sum of money invested at compound interest amounts to 800 in 3 years and to 840 in 4 years. What is the rate of interest per annum? 1) 2.5% 2) 4% 3) 5% 4) 6.66% 5) None of these

$\frac{840}{800} \frac{21}{20} =$

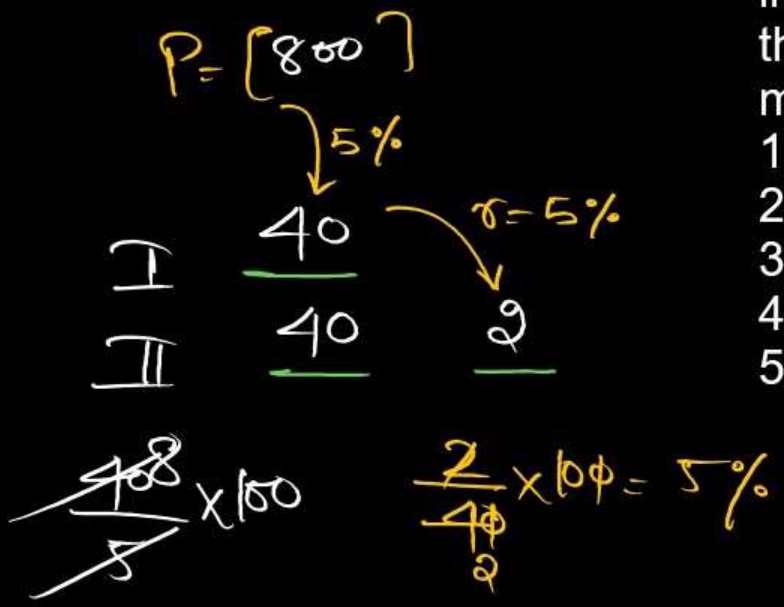
$20 \longrightarrow 21$

$\frac{1}{20} \times 100 = 5\%$

Question 16:

On a particular amount, the compound interest at the end of one year is 40 and in the 2nd year is 42. How much money was deposited?

1) 850
2) 900
3) 800
4) Cannot be determined
5) None of these



Question 17:

What is the difference between compound interest and simple interest for the sum of 20000 over a 2 year period, if the compound interest is calculated at 20% p.a. and simple interest is calculated at 23% p.a.?

1) 1)200
2) 2) 125
3) 3) 250
4) 4) 400
5) 5) None of these

If the CI on a certain sum for 2 years at 5% be Rs 410, what would be the simple Interest ?

TITA

If the CI on a certain sum for 2 years at 4% be Rs 510, what would be the SI?

TITA

The compound interest on a certain sum of money for 2 years at 10% per annum is Rs 420. The simple interest on the same sum at the same rate and same time will be: